

Experiment 25: Perform Sharpening of Image using Gradient Masking

AIM

To sharpen an image using gradient masking (Sobel operator).

PROCEDURE

1. Read the image.
2. Convert it to grayscale.
3. Apply Sobel X and Sobel Y operators.
4. Combine the gradients.
5. Display the output.

PROGRAM

```
import cv2

# Read image

img = cv2.imread(r"C:\Users\chait\OneDrive\Documents\Desktop\cv lab image.jpg")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Sobel Gradient

sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobely = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

gradient = cv2.addWeighted(sobelx, 0.5, sobely, 0.5, 0)

cv2.imshow("Original Image", img)
cv2.imshow("Gradient Mask", gradient)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT

The image sharpened using gradient masking is displayed.

Explanation



We applied Sobel gradient operators along both X and Y directions and combined them. This highlights edges and enhances image details using gradient information.